

# Member Contributions — Group 52 (Super Mario Game)

**Course:** CS202 - C++ Programming / Object-Oriented Design

**Class:** 25A01

**Group:** 52

This document is the same data as [Member\\_Contributions.xlsx](#) (the TA-facing spreadsheet, in the format used by the group's other CS202 project) rendered as Markdown/PDF for readability — both are generated from one source so they cannot drift apart.

## Summary

|                        |       |
|------------------------|-------|
| Number of students     | 2     |
| Number of tasks        | 118   |
| Number of task hours   | 214.5 |
| Number of Git commits  | 316   |
| Max student percentage | 0.5   |
| Project score          | 10    |

*Do not edit the grey cells in the spreadsheet edition. TAs enter the score of all PAs in the project score and get the individual scores; students may enter an estimated project score to see how percentages affect it. `Percent` is set to 0.5/0.5 for both members — the raw Tasks/Hours/Git percentages below inform that judgment rather than dictate it, the same convention the reference spreadsheet uses.*

| No | Student ID | Full name   | Tasks | Tasks % | Task Hours | Hours % | Git Commits | Git % | Percent | Score (Student) |
|----|------------|-------------|-------|---------|------------|---------|-------------|-------|---------|-----------------|
| 1  | 25125083   | Nguyễn Đình | 73    | 61.9%   | 134.5      | 62.7%   | 258         | 81.6% | 0.5     | 10              |

| No | Student ID | Full name    | Tasks | Tasks % | Task Hours | Hours % | Git Commits | Git % | Percent | Score (Student) |
|----|------------|--------------|-------|---------|------------|---------|-------------|-------|---------|-----------------|
|    |            | Minh Huy     |       |         |            |         |             |       |         |                 |
| 2  | 25125084   | Trần Gia Huy | 45    | 38.1%   | 80.0       | 37.3%   | 58          | 18.4% | 0.5     | 10              |

Note on the gap between the Git % column and the Tasks/Hours % columns: Member A's git identity carries a large share of small fix/doc/log commits (258 of 316) accumulated across the whole session history, while task count and estimated hours — a better proxy for actual design/implementation effort — split closer to 62/38. Both are shown rather than only the more flattering one.

## Task-by-task breakdown

A task should be done by only 1-2 students. Each member worked far more than the minimum 5 tasks the reference template asks for.

| No | Student ID | Full name            | Task Description                                                                                                          | Hours | Evidence                      |
|----|------------|----------------------|---------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------|
| 1  | 25125083   | Nguyễn Đình Minh Huy | Configured CMake (FetchContent) for SFML 3.0.2 + Dear ImGui/ImGui-SFML and set up the project directory structure         | 2     | see docs/<br>Group52_04/52.md |
| 2  | 25125083   | Nguyễn Đình Minh Huy | Drafted the v2.0 project spec (110 features), UML class diagrams and the implementation plan (TASKS.md, TASK_DIVISION.md) | 3     | see docs/<br>Group52_04/52.md |
| 3  | 25125083   | Nguyễn Đình Minh Huy | Wrote clean C++ base skeletons for managers, physics and entities                                                         | 1.5   | see docs/<br>Group52_04/52.md |
| 4  | 25125083   | Nguyễn Đình Minh Huy | Implemented MathUtils (vector ops, clamp, interpolation helpers)                                                          | 1     | see docs/<br>Group52_04/52.md |

| No | Student ID | Full name            | Task Description                                                                                                   | Hours | Evidence                      |
|----|------------|----------------------|--------------------------------------------------------------------------------------------------------------------|-------|-------------------------------|
| 5  | 25125083   | Nguyễn Đình Minh Huy | Built GameStateManager/IGameState state-stack pattern and the Game singleton main loop at a fixed 60Hz timestep    | 3     | see docs/<br>Group52_04/52.md |
| 6  | 25125083   | Nguyễn Đình Minh Huy | Implemented AABB collision primitives and a dynamic SpatialHash broadphase grid                                    | 2.5   | see docs/<br>Group52_04/52.md |
| 7  | 25125083   | Nguyễn Đình Minh Huy | Built the PhysicsEngine update pipeline (gravity, gravity zones/water viscosity, axis-separated tile resolution)   | 2.5   | see docs/<br>Group52_04/52.md |
| 8  | 25125083   | Nguyễn Đình Minh Huy | Implemented CollisionDetector/ CollisionResolver (solid tiles, conveyors, stomps/damage, player-vs-player bounces) | 3     | see docs/<br>Group52_04/52.md |
| 9  | 25125083   | Nguyễn Đình Minh Huy | Restructured Entity attributes to protected with friend-class access and read-only getters                         | 1     | see docs/<br>Group52_04/52.md |
| 10 | 25125083   | Nguyễn Đình Minh Huy | Implemented the IPlayerState decorator pattern (StarDecorator, MegaDecorator) with timers                          | 2     | see docs/<br>Group52_04/52.md |
| 11 | 25125083   | Nguyễn Đình Minh Huy | Coded concrete playable characters (Mario, Luigi, Toad, Peach) with custom physics/abilities                       | 2.5   | see docs/<br>Group52_04/52.md |
| 12 | 25125083   | Nguyễn Đình Minh Huy | Built the Item base class and 12 concrete item subclasses (Mushroom, FireFlower, Coin, Star, etc.)                 | 3     | see docs/<br>Group52_04/52.md |
| 13 | 25125084   | Trần Gia Huy         | Built InputManager (Command Pattern) with dual-player key mapping (P1 WASD/Space/F, P2 Arrows/M)                   | 2     | see docs/<br>Group52_04/52.md |

| No | Student ID | Full name            | Task Description                                                                                                | Hours | Evidence                      |
|----|------------|----------------------|-----------------------------------------------------------------------------------------------------------------|-------|-------------------------------|
| 14 | 25125084   | Trần Gia Huy         | Implemented SoundManager (Singleton) with music/SFX channels, WAV/OGG loading                                   | 1.5   | see docs/<br>Group52_04/52.md |
| 15 | 25125084   | Trần Gia Huy         | Researched/designed the IMovementStrategy interface and 8 enemy AI strategies                                   | 2     | see docs/<br>Group52_04/52.md |
| 16 | 25125084   | Trần Gia Huy         | Prepared abstract skeletons for the Enemy and Block type hierarchies                                            | 1     | see docs/<br>Group52_04/52.md |
| 17 | 25125084   | Trần Gia Huy         | Designed the EntityFactory registry blueprint for JSON-driven instantiation                                     | 1     | see docs/<br>Group52_04/52.md |
| 18 | 25125083   | Nguyễn Đình Minh Huy | Implemented the full Camera class (LERP tracking, screen shake, viewport clamping)                              | 2     | see docs/<br>Group52_05/52.md |
| 19 | 25125083   | Nguyễn Đình Minh Huy | Refactored move/run commands to intent flags on Character/Player instead of direct velocity changes             | 1.5   | see docs/<br>Group52_05/52.md |
| 20 | 25125083   | Nguyễn Đình Minh Huy | Centralized friction/acceleration/run-speed logic into PhysicsEngine::update() with surface-specific properties | 2     | see docs/<br>Group52_05/52.md |
| 21 | 25125083   | Nguyễn Đình Minh Huy | Integrated Camera into PlayingState with dual-view (world vs HUD) rendering                                     | 1.5   | see docs/<br>Group52_05/52.md |
| 22 | 25125083   | Nguyễn Đình Minh Huy | Added mid-air wall-slide friction with a capped slide speed (WALL_SLIDE_SPEED)                                  | 1.5   | see docs/<br>Group52_05/52.md |
| 23 | 25125083   | Nguyễn Đình          |                                                                                                                 | 2.5   | see docs/<br>Group52_05/52.md |

| No | Student ID | Full name            | Task Description                                                                                       | Hours | Evidence                      |
|----|------------|----------------------|--------------------------------------------------------------------------------------------------------|-------|-------------------------------|
|    |            | Minh Huy             | Integrated nlohmann/json and built LevelLoader to parse tile/entity/spawn data from level JSON         |       |                               |
| 24 | 25125083   | Nguyễn Đình Minh Huy | Authored 4 level files (level_1..3, bonus_1) and an ImGui level selector for playtesting               | 2     | see docs/<br>Group52_05/52.md |
| 25 | 25125084   | Trần Gia Huy         | Implemented IMovementStrategy and 8 concrete AI strategies (Patrol, Chase, Fly, etc.)                  | 2.5   | see docs/<br>Group52_05/52.md |
| 26 | 25125084   | Trần Gia Huy         | Coded the Enemy base class plus Goomba, KoopaTroopa, KoopaParatroopa, Boo                              | 3     | see docs/<br>Group52_05/52.md |
| 27 | 25125084   | Trần Gia Huy         | Coded the Block base class plus BrickBlock, QuestionBlock, Pipe, Flagpole                              | 2.5   | see docs/<br>Group52_05/52.md |
| 28 | 25125084   | Trần Gia Huy         | Coded 7 advanced v2.0 enemies (PiranhaPlant, BulletBill, HammerBro, Thwomp, ChainChomp, Lakitu, Spiny) | 3.5   | see docs/<br>Group52_05/52.md |
| 29 | 25125084   | Trần Gia Huy         | Coded 5 advanced v2.0 blocks (HiddenBlock, MovingPlatform, FallingPlatform, IceBlock, ConveyorBelt)    | 2.5   | see docs/<br>Group52_05/52.md |
| 30 | 25125084   | Trần Gia Huy         | Implemented EntityFactory to instantiate 25+ entity types from JSON                                    | 2.5   | see docs/<br>Group52_05/52.md |
| 31 | 25125084   | Trần Gia Huy         | Built SpriteSheet, Animation (flyweight) and AnimationManager/Animator                                 | 3     | see docs/<br>Group52_05/52.md |
| 32 | 25125084   | Trần Gia Huy         | Wrote verify_enemies_new.cpp / verify_blocks_new.cpp test harnesses                                    | 1.5   | see docs/<br>Group52_05/52.md |
| 33 | 25125083   | Nguyễn Đình          |                                                                                                        | 2.5   | see docs/<br>Group52_06/52.md |

| No | Student ID | Full name            | Task Description                                                                                     | Hours | Evidence                      |
|----|------------|----------------------|------------------------------------------------------------------------------------------------------|-------|-------------------------------|
|    |            | Minh Huy             | Built Serializer for save-slot JSON schemas (character, player state, scores, checkpoints, playtime) |       |                               |
| 34 | 25125083   | Nguyễn Đình Minh Huy | Implemented StatisticsTracker (EventBus subscriber for coins/enemies/deaths/combos/playtime)         | 1.5   | see docs/<br>Group52_06/52.md |
| 35 | 25125083   | Nguyễn Đình Minh Huy | Implemented AchievementManager (10 achievements) plus a slide-in/out toast HUD                       | 2     | see docs/<br>Group52_06/52.md |
| 36 | 25125083   | Nguyễn Đình Minh Huy | Built the ImGui Pause/Settings panel (save slots, volume, keybindings, difficulty, auto-save)        | 2     | see docs/<br>Group52_06/52.md |
| 37 | 25125083   | Nguyễn Đình Minh Huy | Wrote the verify_save_load.cpp integration test suite                                                | 1     | see docs/<br>Group52_06/52.md |
| 38 | 25125083   | Nguyễn Đình Minh Huy | Implemented the Undo/Redo command framework (EditorCommands)                                         | 2     | see docs/<br>Group52_06/52.md |
| 39 | 25125083   | Nguyễn Đình Minh Huy | Built the MapEditor viewport dashboard (tile/entity brushes, undo/redo, save)                        | 2     | see docs/<br>Group52_06/52.md |
| 40 | 25125083   | Nguyễn Đình Minh Huy | Implemented pixel-to-grid coordinate mapping and grid-line rendering                                 | 1.5   | see docs/<br>Group52_06/52.md |
| 41 | 25125083   | Nguyễn Đình Minh Huy | Centralized duplicate tile/entity name mappers into SerializationUtils                               | 1     | see docs/<br>Group52_06/52.md |

| No | Student ID | Full name            | Task Description                                                                                                                                     | Hours | Evidence                      |
|----|------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------|
| 42 | 25125083   | Nguyễn Đình Minh Huy | Wrote the verify_map_editor.cpp test harness                                                                                                         | 1     | see docs/<br>Group52_06/52.md |
| 43 | 25125084   | Trần Gia Huy         | Participated in architectural reviews, resolved cross-branch merge conflicts, refined achievement SFX triggers                                       | 1     | see docs/<br>Group52_06/52.md |
| 44 | 25125083   | Nguyễn Đình Minh Huy | Merged A/feature/save-load and A/feature/level-editor; addressed PR review fixes (EventBus sentinel, debug key remap, Game keybinding encapsulation) | 2     | see docs/<br>Group52_07/52.md |
| 45 | 25125083   | Nguyễn Đình Minh Huy | Fixed a stale achievement-toast bug on slot reload; aligned achievement keys with SPEC.md                                                            | 1     | see docs/<br>Group52_07/52.md |
| 46 | 25125083   | Nguyễn Đình Minh Huy | Built the consolidated verify_all.cpp test runner combining 6 test suites; updated CMakeLists.txt                                                    | 2     | see docs/<br>Group52_07/52.md |
| 47 | 25125083   | Nguyễn Đình Minh Huy | Authored two_player_ai_plan.md (2-player + Shadow Mario architecture design doc)                                                                     | 2.5   | see docs/<br>Group52_07/52.md |
| 48 | 25125083   | Nguyễn Đình Minh Huy | Ran an interactive binary demo to validate save/load and level-editor integration                                                                    | 1     | see docs/<br>Group52_07/52.md |
| 49 | 25125084   | Trần Gia Huy         | Entity Factory & concrete entities merged into dev (integration of prior work)                                                                       | 0.5   | see docs/<br>Group52_07/52.md |
| 50 | 25125084   | Trần Gia Huy         | Enemy hierarchy + AI strategies merged into dev (integration of prior work)                                                                          | 0.5   | see docs/<br>Group52_07/52.md |

| No | Student ID | Full name            | Task Description                                                                                                                                           | Hours | Evidence                      |
|----|------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------|
| 51 | 25125084   | Trần Gia Huy         | Blocks/platforms merged into dev (integration of prior work)                                                                                               | 0.5   | see docs/<br>Group52_07/52.md |
| 52 | 25125084   | Trần Gia Huy         | Built Hud (lives/score/coins/time/ stage) plus a boss health bar and retro typography (PressStart2P/ SuperMario256 fonts)                                  | 2     | see docs/<br>Group52_07/52.md |
| 53 | 25125084   | Trần Gia Huy         | Finalized SpriteSheet/ AnimationManager/Animator; added verify_blocks/verify_enemies/ verify_graphics/verify_graphics_visual/ verify_hud_boss test targets | 1.5   | see docs/<br>Group52_07/52.md |
| 54 | 25125084   | Trần Gia Huy         | Consolidated duplicate asset/ and assets/ folders into one assets/ root                                                                                    | 0.5   | see docs/<br>Group52_07/52.md |
| 55 | 25125084   | Trần Gia Huy         | Built the Minimap component (scaled tilemap preview, camera frustum, player tracking dot)                                                                  | 2     | Git commit<br>cb6eab2         |
| 56 | 25125084   | Trần Gia Huy         | Wrote the verify_minimap_visual.cpp test target                                                                                                            | 1     | see docs/<br>Group52_07/52.md |
| 57 | 25125083   | Nguyễn Đình Minh Huy | Reviewed/validated the ParticleSystem/ObjectPool design (O(1) acquire/release, zero-allocation)                                                            | 1     | see docs/<br>Group52_08/52.md |
| 58 | 25125083   | Nguyễn Đình Minh Huy | Updated TASKS.md through Phase 8 and Bonus A, cross-referenced against TASK_DIVISION.md                                                                    | 0.5   | see docs/<br>Group52_08/52.md |
| 59 | 25125083   | Nguyễn Đình Minh Huy | Ran git fetch --all, verified dev in sync with origin/dev, tracked branch status                                                                           | 0.5   | see docs/<br>Group52_08/52.md |
| 60 | 25125084   |                      |                                                                                                                                                            | 2.5   |                               |



| No | Student ID | Full name            | Task Description                                                                                                  | Hours | Evidence                              |
|----|------------|----------------------|-------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------|
|    |            | Trần Gia Huy         | Implemented ParticleSystem with ObjectPool (burst/continuous emission modes)                                      |       | Git commit 7a6f62b                    |
| 61 | 25125084   | Trần Gia Huy         | Tuned particle parameters (alpha fade, emission rate, spread angle)                                               | 1.5   | Git commits c7d4fd4, 62c78ef          |
| 62 | 25125084   | Trần Gia Huy         | Built the verify_particles_visual interactive test target                                                         | 1     | see docs/Group52_08/52.md             |
| 63 | 25125084   | Trần Gia Huy         | Extracted the player sprite sheet player.json (92 animation frames: idle/walk/run/jump/etc.)                      | 3     | Git commits d6bd07d, 0be4508, 9403be5 |
| 64 | 25125084   | Trần Gia Huy         | Built the verify_player_animation_visual test target                                                              | 1     | see docs/Group52_08/52.md             |
| 65 | 25125084   | Trần Gia Huy         | Began sprite naming standardization / asset pipeline scripts                                                      | 1     | Git commit 15327b4                    |
| 66 | 25125084   | Trần Gia Huy         | Enhanced random-number generation with std::uniform_real_distribution for particle spread                         | 0.5   | Git commit 9bb9d16                    |
| 67 | 25125083   | Nguyễn Đình Minh Huy | Implemented the Fireball entity + ObjectPool, EntityFactory/EventBus wiring, verify_fireball.cpp (4/4 pass)       | 3     | see docs/Group52_09/52.md             |
| 68 | 25125083   | Nguyễn Đình Minh Huy | Enhanced Flagpole with height-scaled scoring, animation and a LevelComplete event; verify_flagpole_trampoline.cpp | 2     | see docs/Group52_09/52.md             |
| 69 | 25125083   | Nguyễn Đình Minh Huy | Enhanced Trampoline bounce mechanics (spring compression, configurable impulse)                                   | 1.5   | see docs/Group52_09/52.md             |
| 70 | 25125083   |                      |                                                                                                                   | 2     |                                       |

| No | Student ID | Full name            | Task Description                                                                                                                       | Hours | Evidence                      |
|----|------------|----------------------|----------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------|
|    |            | Nguyễn Đình Minh Huy | Built the CheckpointFlag entity (visual states, auto-save, CheckpointActivated event, respawn coordinates)                             |       | see docs/<br>Group52_09/52.md |
| 71 | 25125083   | Nguyễn Đình Minh Huy | Implemented the Memento pattern: GameSnapshot + TimeRewindManager (300-frame circular buffer);<br>verify_memento_rewind.cpp (4/4 pass) | 3.5   | see docs/<br>Group52_09/52.md |
| 72 | 25125083   | Nguyễn Đình Minh Huy | Built MapGenerator for procedural level generation;<br>verify_map_generator.cpp                                                        | 3     | see docs/<br>Group52_09/52.md |
| 73 | 25125083   | Nguyễn Đình Minh Huy | Implemented the dual bounding-box architecture (physicsBox/combatBox) across Entity/CollisionDetector/CollisionResolver                | 2.5   | see docs/<br>Group52_09/52.md |
| 74 | 25125083   | Nguyễn Đình Minh Huy | Added void-fall death detection in PlayingState                                                                                        | 1     | see docs/<br>Group52_09/52.md |
| 75 | 25125083   | Nguyễn Đình Minh Huy | Added brick/question-block head-butt reactions in PhysicsEngine                                                                        | 1.5   | see docs/<br>Group52_09/52.md |
| 76 | 25125083   | Nguyễn Đình Minh Huy | Removed public setLives/setCoins/setScore from Player; added friend class PlayingState                                                 | 0.5   | see docs/<br>Group52_09/52.md |
| 77 | 25125083   | Nguyễn Đình Minh Huy | Added Camera::move() for editor WASD/arrow panning                                                                                     | 1     | see docs/<br>Group52_09/52.md |
| 78 | 25125083   | Nguyễn Đình          | Added a Level Editor Mode (F1) button to MenuState                                                                                     | 0.5   | see docs/<br>Group52_09/52.md |

| No | Student ID | Full name            | Task Description                                                                                                          | Hours | Evidence                              |
|----|------------|----------------------|---------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------|
|    |            | Minh Huy             |                                                                                                                           |       |                                       |
| 79 | 25125083   | Nguyễn Đình Minh Huy | Hardened Game::shutdown() to pop all game states before manager teardown                                                  | 1     | see docs/Group52_09/52.md             |
| 80 | 25125083   | Nguyễn Đình Minh Huy | Added multi-path font-loading fallback candidates in PlayingState                                                         | 0.5   | see docs/Group52_09/52.md             |
| 81 | 25125083   | Nguyễn Đình Minh Huy | Fetches/merged origin/B/feature/ graphic-visual into dev; resolved font-loading + agent_history.log conflicts; pushed dev | 1     | Git commit 17cef23                    |
| 82 | 25125083   | Nguyễn Đình Minh Huy | Fixed non-solid TileType::Coin tiles not being collected on walk-through                                                  | 1     | Git commit d12e763                    |
| 83 | 25125083   | Nguyễn Đình Minh Huy | Deduplicated a ResourceManager missing-SoundBuffer warning spam                                                           | 0.5   | see docs/Group52_09/52.md             |
| 84 | 25125084   | Trần Gia Huy         | Built a spritesheet-merging tool; updated enemy_projectile/item/world_scenery_item atlases                                | 3     | Git commits 044a5d8, 80c048e, b90a899 |
| 85 | 25125084   | Trần Gia Huy         | Cleaned sprite/atlas coordinates affected by the merge                                                                    | 1     | Git commit 10828a7                    |
| 86 | 25125084   | Trần Gia Huy         | Updated SPRITE_NAMING_CONVENTIONS.md (frame inventories, naming standards)                                                | 1     | see docs/Group52_09/52.md             |
| 87 | 25125084   | Trần Gia Huy         | Built SpriteColorFilter (HSL->RGB, 12Hz Star Power cycling, 15Hz hurt flicker, hit-flash tint)                            | 2     | Git commit 2c3bb83                    |

| No | Student ID | Full name            | Task Description                                                                                                                                                                              | Hours | Evidence                  |
|----|------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------|
| 88 | 25125084   | Trần Gia Huy         | Built SpriteTransformAnim (scale-lerp easing, 720 deg/s spin, 12Hz Mega/Mini flicker)                                                                                                         | 2     | Git commit 2c3bb83        |
| 89 | 25125084   | Trần Gia Huy         | Built the EntityDeathEffect singleton (EnemyFlip, StarKillSpin, PlayerDeathHop)                                                                                                               | 2     | Git commit 2c3bb83        |
| 90 | 25125084   | Trần Gia Huy         | Added a SoundManager synthesized fallback beep buffer                                                                                                                                         | 1     | see docs/Group52_09/52.md |
| 91 | 25125084   | Trần Gia Huy         | Upgraded verify_graphics_visual with a Transform/Color-Filter FX ImGui tab                                                                                                                    | 1.5   | see docs/Group52_09/52.md |
| 92 | 25125083   | Nguyễn Đình Minh Huy | Refined MapGenerator (elevation profiles, biome ceilings, lava pits, jump-reachability-guarded platforms, 3 Star Coins, threat-pacing curve) + ImGui tuning panel                             | 3.5   | see docs/Group52_10/52.md |
| 93 | 25125083   | Nguyễn Đình Minh Huy | Built the 3-world campaign + warp-pipe system; generate_game_levels.cpp synthesizing 7 level JSONs; extended SerializationUtils/LevelLoader for warp metadata; seamless sub-level transitions | 4     | see docs/Group52_10/52.md |
| 94 | 25125083   | Nguyễn Đình Minh Huy | Added ImGui campaign navigation dropdowns and an Active Level Tube warp selector                                                                                                              | 1.5   | see docs/Group52_10/52.md |
| 95 | 25125083   | Nguyễn Đình Minh Huy | Fixed a static-block gravity bug (getGravityMultiplier() override); added item resting-state physics (resolveItemVsBlock)                                                                     | 1.5   | see docs/Group52_10/52.md |
| 96 | 25125083   | Nguyễn Đình          | Optimized player hitbox dimensions (32px to 24px width tiers); refined stomp-vs-side-hit combat resolution                                                                                    | 2     | see docs/Group52_10/52.md |

| No  | Student ID | Full name            | Task Description                                                                                                                                                                          | Hours | Evidence                                                                  |
|-----|------------|----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------|
|     |            | Minh Huy             |                                                                                                                                                                                           |       |                                                                           |
| 97  | 25125083   | Nguyễn Đình Minh Huy | Fixed the LevelLoader ifstream failbit cascade (file.clear()); expanded fallback search paths                                                                                             | 1.5   | see docs/<br>Group52_10/52.md                                             |
| 98  | 25125083   | Nguyễn Đình Minh Huy | Fixed a SIGABRT shutdown crash (explicit m_window.close()) and hardened GameStateManager transitions                                                                                      | 1.5   | see docs/<br>Group52_10/52.md                                             |
| 99  | 25125084   | Trần Gia Huy         | Wired sprite animations across all entities; aspect-ratio scaling for HUD/items; POW 8-frame shockwave; trampoline/PSwitch visual states                                                  | 3.5   | Git commits<br>069817b, 0bc21c3,<br>2d5058e, 83660b5,<br>065557d, 61e30bc |
| 100 | 25125084   | Trần Gia Huy         | Built the Web Metadata Editor Studio (metadata_editor.html, edit_sprite_metadata.py) with cursor-anchored zoom, position-finder HUD, drag-to-draw frames, TexturePacker-compatible export | 2.5   | see docs/<br>Group52_10/52.md                                             |
| 101 | 25125084   | Trần Gia Huy         | Integrated sprite HUD assets (animated coin counter, Star Coin indicators, character-specific lives icon)                                                                                 | 1.5   | see docs/<br>Group52_10/52.md                                             |
| 102 | 25125084   | Trần Gia Huy         | Built PipeRenderer (2-block-wide AABB, directional multi-part pipe head/body rendering)                                                                                                   | 1.5   | Git commit<br>a0f1a3b                                                     |
| 103 | 25125084   | Trần Gia Huy         | Wired game-wide SFX/BGM into SoundManager (jump, stomp, coin, break, power-up/down, boing, stage-clear, game-over) + fallback buffer                                                      | 2.5   | Git commits<br>1002595, c383c19,<br>9976209                               |
| 104 | 25125084   | Trần Gia Huy         | Tuned all 11 enemy AI state machines (Thwomp slam cycle, Boo gaze-detection, HammerBro parabolic throw, Koopa shell physics)                                                              | 3     | Git commit<br>d954a66                                                     |

| No  | Student ID | Full name            | Task Description                                                                                                                        | Hours | Evidence                                        |
|-----|------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------|
| 105 | 25125084   | Trần Gia Huy         | Built verify_enemies_behavior.cpp + enhanced verify_enemies_visual.cpp (11 isolated test rooms)                                         | 1.5   | Git commit d954a66                              |
| 106 | 25125083   | Nguyễn Đình Minh Huy | Resolved the Level 1-3 crash on Bowser's first fireball (mid-frame entity-spawn vector reallocation); rebuilt and playtested on Windows | 3     | see logs/<br>agent_history.log<br>2026-08-16/22 |
| 107 | 25125083   | Nguyễn Đình Minh Huy | Fixed 12 defects from a playtest pass plus two menus that overflowed                                                                    | 2     | Git commit f6fc8bc                              |
| 108 | 25125083   | Nguyễn Đình Minh Huy | Fixed CMakeLists naming src/Graphics/HUD.cpp when the file is Hud.cpp                                                                   | 0.5   | Git commit 03bf828                              |
| 109 | 25125083   | Nguyễn Đình Minh Huy | Fixed the player being unable to jump while walking into a wall                                                                         | 1     | Git commit 60cc735                              |
| 110 | 25125083   | Nguyễn Đình Minh Huy | Fixed 28 files using std:: symbols without including their header                                                                       | 1     | Git commit 4ac8a1c                              |
| 111 | 25125083   | Nguyễn Đình Minh Huy | Fixed CI building only 2 of the 13 binaries it then tested                                                                              | 1     | Git commit 8141120                              |
| 112 | 25125083   | Nguyễn Đình Minh Huy | Fixed 4 test harnesses opening an sf::RenderWindow on a CI runner with no display                                                       | 1     | Git commit 33d5aae                              |
| 113 | 25125083   | Nguyễn Đình Minh Huy | Fixed three defects the screenshots showed that the tests did not (backdrop ground line, hill-bottom                                    | 2     | Git commit da8ca99                              |

| No  | Student ID | Full name            | Task Description                                                                                                                                                                                                                               | Hours | Evidence                                                       |
|-----|------------|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------|
|     |            |                      | offset, a Player 2 HUD badge overwriting the world field)                                                                                                                                                                                      |       |                                                                |
| 114 | 25125083   | Nguyễn Đình Minh Huy | Sealed test hermeticity: Serializer::setSaveDirectory() + TestSaveSandbox.hpp so all 23 verify_* harnesses point at a fresh temp directory                                                                                                     | 2.5   | Git commit 045f9b7                                             |
| 115 | 25125083   | Nguyễn Đình Minh Huy | Generated real UML class diagrams from the headers (tools/gen_class_diagram.py) and wrote the CS202 final report's first edition (15 sections, generated by build_report.py)                                                                   | 4     | Git commits a0395d6, 3c0a768, 64088b4                          |
| 116 | 25125083   | Nguyễn Đình Minh Huy | Produced a LaTeX/PDF edition of the report in the HCMUS template and deepened it (implementation section 3->11 subsections, technical-problems section rebuilt into 8 themed groups)                                                           | 4     | Git commits fdec9cc, 730336c                                   |
| 117 | 25125083   | Nguyễn Đình Minh Huy | Diagnosed and fixed three defects reported from a real playthrough: the end-of-level castle's buffer margin, Boom Boom's arena overlapping the 1-2 flagpole, and the camera/z-order bug carried into 1-3; added regression tests for all three | 4     | this session, branch A/fix/level-completion-and-camera-defects |
| 118 | 25125084   | Trần Gia Huy         | Resolved merge conflicts while pulling the latest changes and merged into main                                                                                                                                                                 | 2     | Git commits 72b13c6, a522007, 75e9cbb                          |